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Chapter 5
Scientific Tarot: Reading Hands and Strategies

Players are often fond, especially on television and in interviews, of the concept of "reading" opponents and the idea that they do this better than their opposition. Implicit in many of the characterizations of the process of "reading" is the idea that a "read" cannot be quantified and mat unimaginative "math guys" somehow lack this ability. Unsurprisingly, we disagree. While the ability to accurately pinpoint the hand an opponent holds by "magic" would be in fact powerful, it is our view that reading opponents is largely a process of Bayesian inference -elated to betting patterns, with some 'normally small) adjustments made for physical tells. This is true whether the process of inference is explicit or subconscious. Using a mathematical model can greatly enhance one's intuition as a guide to proper play.

One popular method of "reading" opponents involves guessing at our opponent's hand, playing as if he held that hand, and hoping to be right. Sometimes this leads to the correct play for the situation. One example of this type of thinking are players who reraise early raisers in limit holdem with small to medium pairs in position, "putting their opponents on AX." It is true that AK. is (assuming the opponent will always raise this hand) the most likely single -and that an early raiser can hold; so frequently this "read" will turn out to be correct. Nonetheless, these players could be sacrificing equity both preflop and postflop by assuming mat theft opponent holds a specific hand, as his distribution contains not only unpaired big cards, but also larger pairs.

One of the authors recently played a hand in a no-limit satellite that was illustrative of this point. There were five players left, the blinds were 150-300, and a player who had pushed his short stack all-in several times in the last ten or so hands raised all-in to about 2100 from the under-the-gun position. It was folded around to the author in the big blind, who had about 3200 in chips and held A♥ 8♥. The author called. The raiser held AKo, and the author caught an eight and won the hand.

After the satellite was over, the player who held AKo questioned the author, "What did you think I had that you were ahead?" The author responded, "Well, I figured you were jamming there with any ace. any pair, and some good kings. Plus, the stronger you are as a player, the more likely you are raising with hands like T9s and the like. Against that range. I had almost 50% equity, so I think it's an easy call."

The point of the story is that the questioner was trying to imagine the author's thought process as "putting him on a hand," and then evaluating his chances against that particular hand. But no such process occurred. Instead, your author created a distribution of hands with which the player could be raising and acted accordingly. As it happened, the player held one of the stronger hands in his distribution and the author was lucky to win the hand. But the question of "being ahead" of a specific hand was never part of the decision process for the *author, and the call was* still *fundamentally correct.*

Advanced Hand Reading

Other players "read" their opponents by using a combination of logical deduction and tells. These players rule out various options based on assumptions that theft opponents play reasonably or that they have an idea of how their opponents play. One common error made by those who practice this type of hand reading is to strip hands from the range of the opponent too aggressively and thus improperly narrow the distribution of hands the opponent might hold.

Nevertheless, what these players do is much more akin to the hand-reading process that we

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advocate; in its most sophisticated forms it is very much the same. In theory, the approach is as follows: We generally never assign one single hand to an opponent; instead, each opponent has a probability distribution of different possible hands. At the beginning of a hand, each opponent has a full distribution of random hands, adjusted for the Bayesian card-removal effects of our own hand. Then as each player acts, we adjust for the new information by modifying the various probabilities, both for the new cards revealed, and for our best estimates of what actions the players would take with each hand. We often include an auxiliary probability, which we might call a "lost his mind" probability, which reflects the (sometimes small) probability that our understanding of the player's style is incomplete or that he is simply deviating from his usual strategy. Players, even fairly strong ones, occasionally do things that are quite out of the ordinary, and pretending that these things are probability zero is simply incorrect and can lead to making incorrect "big laydowns" or bad calls and raises.

In practice, of course, we do not hold all the exact probabilities for each hand in our head; this would be overwhelming and much more trouble that it would be worth in terms of making decisions. However, we can usually reconstruct the betting and exposed cards in such a way that we can create this distribution for a player at a point where we need to make an important decision. Additionally, it is not difficult to incorporate physical tells into this framework; we simply apply an additional layer of Bayes' theorem to our existing distribution. That is, we ask "Given the probability distribution of hands that this player currently holds, what is the probability that he would exhibit the tell that I picked up on with each one?" and rebalance the probability distribution in fight of this additional information.

The ultimate goal here is to incorporate all the information we can gather to ***find the probability distribution of hands that our opponent holds.*** This is generally a process of reduction and elimination; hands that the opponent would have played in some clearly different way should be reduced in relative probability within the distribution. Ideally, we could gather enough information to find a distribution that contains only one hand. In practice, though, gathering this much information is normally impossible.

A detailed example may illustrate the principles we're describing. In this example, we will make frequent assumptions about the meaning of play on certain streets. In many cases, these are assumptions that can be questioned and debated. The point of this exercise is not so much to argue about the proper way to play a particular hand but to illustrate the process of hand-reading described above. In this section, we will examine the hand-reading aspects of the hand; in a later chapter, we will consider the key decision that appears as the hand unfolds.

Example 5.1

The game is seven-card Stud eight-or-better. The antes are $5, and the limits are $30-60. A player to our right brings it in with the 2♣ for $10. We complete the bet to $30 with (5♣ A♥) 4♥. Two tens, the 7♦. and a king fold. The next player raises to $60 with the 6♣. The bring-in folds, and we call.

(5♣ A♥) 4♥
(??) 6♣

Folded cards: T♣ T♠ ***7♦*** K♠ 2♣.

On third street, we begin to create our picture of the opponent's hand. Without really knowing anything about the player, we can still begin to construct a rough distribution of hands he might hold. Among the hands that would likely merit a re-raise from almost any player:

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(AA) 6
(KK) 6
(66) 6

Three small clubs including the ace.

Our slightly more aggressive opponents' raising distribution would include:

(QQ-TT) 6
Any three small clubs
(A2) 6
(A3) 6

If the opponent were quite aggressive, then we might include:

(99-55) 6
(A4) 6
(A5) 6
(54) 6
(57) 6

Any three clubs with in ace, such as (A♣ J♣) 6♣

And your authors have both seen such re-raises with hands such as:

(QJ) 6

Unfortunately for our hand-reading methodology but fortunately for our bankrolls, the players who make this type of plays often lose their money too quickly to get a truly accurate read on then raising distribution.

At this early stage in the hand, we can only approximate the opponent's distribution. However, it is *worth* noting that certain of these hands are already fairly unlikely. For example, there are only two tens remaining in the deck, so (TT)6 is one-sixth as likely as it might have been had there been no folded cards. Also, a player with (TT)6 would likely see his tens out and fold his hand instead of reraising.

On fourth street, we catch the 8s and the other player catches the T♦. This is a fortunate turn of events. The opponent now checks and we bet, expecting to occasionally pick up the pot right away. However, the opponent calls.

On this street, we pick up very litde information. There are a couple of reasons why we cannot make strong inferences. The first reason is that the betting followed a predictable pattern; we caught a moderately good card, while our opponent caught a fairly bad one. However, the size of the pot and the relative strengths of our distributions (a topic we will return to at length later) make it reasonable for the opponent to flat call pending fifth street. The second is that the betting limits double on the next round. It is often correct for a player ;o defer raising on a smaller street when doing so will give away information and he will likely get to raise on the next street anyway.

On fifth street we catch the J♥. a rather poor card, although it does give us a three-flush in addition to our four low cards. The opponent catches the K♣ and checks again. We now bet,

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expecting to pick up the pot fairly often.

The hands are now:

(5♣A♥) 4♥ 8♠ J♥
(????) 6♣ T♦ K♣

Now let's consider what the implications of our opponent's various actions might be. Looking back to our candidate hands, especially those from the first two categories: We have probable calling hands:

(AA)	6♣ T♦ K♣
(X♣Y♣)	6♣ T♦ K♣
(QQ)	6♣ T♦ K♣
(JJ)	6♣ T♦ K♣

Probably raising hands:

(66)	6♣ T♦ K♣
(KK)	6♣ T♦ K♣

Probably folding hands:

(A2)	6♣ T♦ K♣
(A3)	6♣ T♦ K♣

Note that for each of these categories, we only identify the hands as "probably" indicating some action; it's entirely unclear how our opponents will actually play these hands, or what percentage of the time they will take each action. However, it seems that raising as a semi-bluff with four clubs is the most likely deviation from these strategies. However, other such deviations cannot be ruled out. In creating distributions, it is rather important not to ignore the fact that opponents will play hands in ways we do not anticipate. Ruling out a hand based on prior action can be costly if doing so causes us to make supposedly "safe" raises later in the hand, only to be reraised and shown a better hand.

Now let's say that the opponent calls. We have tentatively narrowed his hand to the big pairs and to four-flushes. On sixth street, we catch the A♠, while the opponent catches the 3♣.

We have a decision to make.

Recapping the hand so far:

(5♣A♠)	4♥	8♠	J♥	A♠
(????)	6♣	T♦	K♣	3♣

There is $345 in the pot at this point. The question that appears here is: should we bet? Many players would send-automatically bet their aces and a low draw. But how strong is this hand given the assumptions we made above? And also importantly, how close is the decision? If we change the assumptions by a small amount, does it change the entire analysis?

To answer these questions, we must examine the opponent's assumed distribution at this point in the hand and our EV against his entire distribution.

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First, let us identify the hands our opponent might hold. For now we will neglect the "lost his mind" category, but once we find an answer we will test it against distributions that include more of those hands.

We classified the opponent's hands into four categories:

(AA)	6♣	T♦	K♣	3♣
(X♣Y♣)	6♣	T♦	K♣	3♣
(QQ)	6♣	T♦	K♣	3♣
(JJ)	6♣	T♦	K♣	3♣

These were based on how he played the hand throughout. Now, however, we must further subdivide the QQ and JJ hands into hands that contain the appropriate club, and hands that do not. (All AA hands contain the ace of clubs, of course-we have the other aces in our hand!). No queens are dead, and of the six possible pairs of queens, three contain the queen of clubs. The jack of hearts is dead (in our hand), so there remain only three pairs of jacks and two of them contain the J♣.
One last step is to figure out how many "three low clubs" hands are possible. We know that the 2♣, 3♣, 5♣, and 6♣ are all outside of the opponent's hole cards, as the 2♣ was folded on third - street, and the other cards appear in our hand or on the opponent's board. There are then four low clubs remaining: A♣, 4♣, 7♣, 8♣. This yields six different possibilities containing two clubs.

So we have the following possibilities, considering just the hole cards:

AA	-1
Q♣ Qx	-3
Qx Qy	-3
J♣ Jx	-2
Jx Jy	-1
X♣ Y♣	-6

In addition to this, there might be additional information available. For example, incorporating a tell that our opponent does not like his hand when the ace hits dearly indicates a *bet if we* think that he would exhibit this tell when he has a big pair in the hole but not otherwise. On the contrary, a tell that he is very pleased to see the 3♣ can be extremely valuable, and might save us at least one full bet if he has made a flush.

To recap, we began with a probability distribution that assigned an equal probability to every pair of hole cards that our opponent could hold. When he raised on third street, we were able to narrow that range to a subset of all hands that contained mainly big pairs, strong low hands, and low three-Hushes. On fourth street, we gained equity in the pot by catching a decent card while the opponent caught badly, but gained very little information.

On fifth street, we gained a great deal of information when he continued even after catching a second high card. It's entirely reasonable that an opponent might take another card off on fourth street with three unsuited low cards after reraising the previous street. However, once the K♣ comes off on fifth, we can effectively rule out all low draw s that do not contain two clubs in the hole. If the opponent does call on fifth with just three low cards, we profit immediately from his mistake in calling. Even if our exploitive process leads to us sacrificing some expectation on later streets, we have already gained enough by his mistake to more than offset the expectation we lose.
On sixth street, our hand improves substantially because we make a high hand; at the same time

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our opponent's distribution improves substantially by catching a small club that is likely to either make him a flush draw or a flush. We narrow his probability distribution down to approximately four classes of hands (aces, big pairs with flush draws, big pairs without flush draws, and made low flushes).

This example primarily made use of the idea of using ***betting patterns*** to read our opponent's hand. However, there is a second important element to reading hands, which is the ability to both spot and correctly interpret tells. As we stated previously, we will omit discussion on how precisely to identify tells that are being exhibited-this falls outside of the scope of this book. However, in the field of correctly interpreting tells, there is much that can be done mathematically. We will return to this topic later in the chapter. We will return to this hand in Chapter 8, at which point we will consider the bet-or-check decision we have on sixth street.

When we play exploitively, we are ***figuring out a distribution of hands and an opponent's strategy*** (or a weighted distribution of strategies) against which to maximize our EV. So far in this chapter, we have discussed the process of "reading hands," which for us is effectively a process of Bayesian inference of hand distributions. This process in turn depended on our estimation of how the opponent would play certain hands. In the examples we looked at previously, we either assumed that opponents would play straightforwardly, or we simply assumed that we knew their strategy to facilitate the analysis.

Reading Strategies

An important and extremely difficult part of playing exploitively is accurately assessing the way in which opponents will play the various hands that are present within their distributions. One possible assumption would be that the opponent will play all the remaining hands in his distribution very well. If we make this assumption and find a corresponding exploitive strategy, that strategy will exploit the shape of his distribution of hands. In cases where the opponent has made a serious error in reaching a given point in the hand, exploiting the shape of his distribution will be quite profitable.

We can additionally profit, however, if our opponents continue to make mistakes in how they play then distribution of hands at decision points to come in the hand. We have several sources of information that are useful in estimating the strategy with which our opponents will play their various hands, a process we will call "strategy reading."

- ***Direct hand evidence***
  Sometimes, we get information about the hand that was played that is known to be truthful because players are forced to show their hands at showdown hi order to win the pot. When it is available, this is perhaps the most valuable type of evidence because many of the other forms of evidence can be falsified in some way.
- ***Direct hand evidence (opponent-controlled)***
  As a special subclass of the above, opponents will sometimes show then hands on purpose. This category is a double-edged sword; we are in fact being shown truthful evidence. However, the fact that this information is being revealed voluntarily by the opponent makes it less valuable because it is likely designed to produce a psychological effect. However, assuming that we are appropriately giving weight to the evidence that we see, we can often use this information to positive effect.
- ***Indirect hand evidence***
  In addition to the hand value information that must be disclosed at showdown, we can also tabulate frequencies of actions, even for hands that are not shown. This information can be useful in both hand and strategy reading, although it must be combined with additional information about the ***types*** of hands that are being played.

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- ***Player classification correlation***
  This is perhaps the most commonly used method of strategy reading both for players about whom we have not gathered much information and for reading strategies generally in situations that come up infrequendy. Essentially this consists of a broad classification of players into different categories or category combinations. Some authors have suggested axes such as loose-tight and passive-aggressive. Others have used animal archetypes as the model for their classifications. The core of all these methods, though, is this. If we can accurately characterize the play of a group of players and then accurately assign a particular player to the proper group, then we can make inferences about a player's play even in situations in which we have never seen him play.

The following discussion will consider primarily players who play reasonably; that is, they play in a manner that is not obviously and easily exploitable, even though they make frequent mistakes. At the end of this discussion, we write about some of the common types of very weak players, whose strategies are much easier to read.

Even though we have these sources of information available, it is important to recognize the imitations of our ability to read our opponents' strategy. It is our view that many players, especially those who are skilled at gathering and retaining the above information, overestimate the degree of certainty with which they can read their opponents' strategies, and as a result - overestimate the predictive value of their player classifications. Even over the course of many hands, we gain relatively little direct evidence of how a player plays.

Consider a sample of 1,000 full ring game limit holdem hands (equivalent to about thirty hours of brick and mortar play and perhaps ten to fifteen hours online). This seems like a fairly large sample of hands with which to "get a line" on an opponent's play. But consider that a typical opponent will likely show down only a hundred of those hands. These hands will be split unevenly amongst the nine table positions with the blinds and the button having a disproportionate share of the total hands shown down. Even further, the hands shown down will be played on quite different textures of flops and against different opponents, in different situations.

Indirect hand evidence is not much more help here, as the sample sizes for preflop actions only slowly reach reliable levels. For example, consider 121 hands played under the gun. If a player raises 10% of hands in the observed sample horn this position, then over the entire sample, a *95%* confidence interval is that he plays roughly between 4% and 16"% of hands. 4% of hands is {TT+, AK, AQs). 16% of hands is (66+, AT+, KJ+, QJ, JT, T9} or some variation. Even seeing a few strong hands that this player raised in this position doesn't allow us to narrow the distribution much, because hands such as QQ are in all the various distributions. In addition, hands like these are more likely to reach the showdown because of their inherent strength. So we have some difficulty in inferring much of value from this indirect evidence as well.

So it seems that the primary source of information about the opponents' play comes from applyhig the direct evidence not to the problem of directly estimating how he will play in various scenarios, but from applying it to the problem of accurately classifying the player and making conclusions about his play based on his classification. Some of the threats to the validity of this method include:

- Players do not employ a single, fixed strategy in the short run: they in turn might be playing exploitively in the observed hand. Using an observed hand as a piece of information toward classifying the opponent relies on the observed hand being characteristic of the opponent's overall strategy. What is often important for exploitive play is not how

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the player plays in general but how the player plays against you. In a later chapter we will discuss counter-exploitation and how the process of opponent adaptation causes problems in exploitive play and eventually leads us to look for other solutions.

- Players do not employ a single, fixed strategy over a longer period; they read books, talk to other players, and come to realizations on their own. Because of this, even a carefully honed read on an opponent's strategy might become invalid in a fairly short period of time. Even such variables as a player winning for a while at a higher limit than he is used to playing might lead him to play more confidently and drastically change his strategy.
- There is comparatively little opportunity to validate and confirm our hypotheses about an opponent's play. For example, many hands are played in a similar manner by many different strategies. If a player plays one of these hands in the typical manner, it only slightly confirms a read on the player's play. It is a rare circumstance where our read that forecasts a ***deviation*** from typical play is confirmed. In order for this to occur, our strategy read must predict a deviation from typical play with a particular hand or hands, the opponent must hold one of those particular hands, and we must somehow get direct information about the hand he held (usually through a showdown). Such an occurrence would be a powerful piece of confirming evidence-but this is rare.

The human mind is quite adept at seeing patterns, even in random sequences. Many studies in the field of psychology have confirmed that people often describe underlying patterns in data presented to them, even when the data is randomly generated and no such pattern exists. As a result of this tendency, we should aggressively seek out information to confirm or contradict our hypotheses. But as we saw above, it is often quite difficult to obtain such information because there are not many chances to observe it.

In this vein, we must also be careful not to assign confirming value to evidence that seems to be ran firming evidence but actually is not. For example, consider the case of a holdem player who folds each hand for the first few orbits, then plays a hand and at the showdown it turns out to be AA. Many players might take this as confirming evidence that this player plays tightly preflop. However, the fact that the player played AA is not confirming evidence of his tightness—in fact, it is only evidence that he was dealt AA. After all, all reasonable strategies would play AA preflop. The thirty hands he folded before ***are*** evidence of his preflop tightness; but the fact that the player played AA is not evidence that he is tight. Of course, it mates for a good story.., "This guy folded thirty hands in a row. Then he played a hand and it's aces." But in fact, the fact that he played aces doesn't add anything to the evidence available.

Our discussion here may give the impression that we are quite negative on the process of reading our opponents' strategies. It is not that we deem this process to be without value; instead, we believe that the high requirement for effort expended, and the sometimes dubious value of the information gained are in sharp contrast to the perceptions of many players, many of whom believe that they can quickly and accurately assess a reasonable player's strategy based on just a few hours at the table with them. The process of characterizing a player's play from the very limited data available often produces a vague sense of how a player might play without the attendant details that make exploitation extremely profitable.

Reading Tells

Many players tend to overvalue their ability to read a hand or spot a physical tell. This frequently occurs because we have selective memories about these types of things; we remember times when we had a dead read on someone's hand, but forget times when we slightly surprised at the showdown. The authors try to train our intuition to accurately assess the value of tells observed. One method is called "the hand reading game." At the showdown of a hand in which we are not involved, just after the river betting concludes, we try to name the hand that one of the

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players in the hand holds. Hitting the hand precisely is a “win,” while missing the hand is a "loss." By doing this, we can get a better sense of how often our reads are accurate and how often we're simply wrong. After doing this exercise for a session, many players are surprised at how infrequently they can actually call then opponent's hands and how often they are incorrect.

With regard to tells, the additional information that we can gain from physical tells in terms of reading hands or strategies is of substantial value. However, we have some difficulties here (which are related to the difficulties we have examined so far). Like historical information, it is difficult to observe repeated confirming instances; there is the same parlay of having an appropriate hand, exhibiting the tell, and then providing confirming evidence in the form of a showdown. However, when we have a confirmed tell, this evidence can be extraordinarily valuable. Often, tells can be applied (in a slightly weakened form) by player classification correlation; that is, we observe a particular tell in a broad group of players, and therefore infer that an unknown player exhibiting the tell is in a similar situation to others who exhibit it.

We can attempt to quantify some of these effects as well. Reading physical tells has much in common with diagnosing a disease, which is a fundamentally Bayesian process. We observe certain effects, and from these effects we attempt to infer the cause. A cold causes a stuffy nose and fatigue; amyotrophic lateral sclerosis causes the progressive loss of motor function. We observe these symptoms and try to ascertain their cause. In the same way, there is a cause and effect relationship between hand strength and tells. The player moves his chips in quickly because he's excited about the strength of his hand; the player holds his breath when bluffing, etc.

The skill here is in inverting the Bayesian network and accurately assessing what the *a priori* distribution looks like in order to properly interpret the new information. Cancer patients report fatigue-but a typical physician would not normally diagnose an otherwise healthy patient with cancer based on that symptom. Nor would his mind leap to that on the small list of possible causes; this is because there are many much more probable causes for fatigue.

Suppose that A is an event, such as "has a bluff," and τ is a tell that we have observed. What we are looking for ultimately is $p(A \mid \mathcal{T})$, the probability that A is true, given that we observe $\mathcal{T}$. We know from Bayes' theorem (Equation 3.1):

$$p(\mathcal{T}|A) = \frac{p(A \cap \mathcal{T})}{p(A)}$$

and also from Equation 1.5:

$$p(A \cap \mathcal{T}) = p(\mathcal{T})p(A|\mathcal{T})$$

Hence,

$$p(\mathcal{T}|A) = \left(\frac{p(\mathcal{T})p(A|\mathcal{T})}{p(A)}\right)$$

$$p(A|\mathcal{T}) = p(A \cap \mathcal{T})/p(\mathcal{T})$$

We also know that $p(A|\mathcal{T}) = p(A \cap \mathcal{T}) + p\,(\overline{A} + \mathcal{T})$ that is, the probability of $\mathcal{T}$ occurring is the probability of A and $\mathcal{T}$ occurring together plus the probability of A not occurring but $\mathcal{T}$ occurring.

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This second term is very important; what it represents is the probability of a ***false positive.*** This is how often we observe this tell and it doesn't have the meaning we ascribe to it. Tells that are very seldom false positives are very valuable, because the probability of A given T approaches 100%. One example of this type of tell is when a player who has not yet acted fails to protect his cards and, for example, turns away from the table. This type of tell is almost never a false positive, because a player with a strong hand would not exhibit this behavior.

Likewise, this leads to the error of a typical tell-reading practice; that is, ascribing meaning to tells that have alternate meanings that contradict the proposed meanings. Suppose we discuss why a player who pushes in his chips quickly might do that. Some arguments can be made on either side of this discussion-the player might have a weak hand and be attempting to look strong: he might be anxious to put his chips in because his hand is strong but vulnerable, and so on. The point is that even if the observation of the tell is very clear- that is, we can be quite sure we have seen this tell exhibited, the tell is of weaker value because the $p\ (\overline{A} + \mathcal{T})$ term is so large that our conditional probability isn't dose to 1 or 0.

The real problem that we want to solve here is a much more complex one. We want to find the probability of some event A, given a sequence of observations that includes some mixture of tells $\mathcal{T}_n$, some observed hands A_n, some unobserved hands, and some observed tell T for the current hand. But we lack many observations with which to accomplish this. Some players' brains seem to do a better job of addressing these conditional probabilities than others-it is likely this effect that causes us to identify players with stronger "reading" abilities.

On a more positive note for exploitive play, there are certain common types of players who play quite poorly about whom we can gain information easily and immediately. Some examples are:

- ***Maniacs*** - We use the term maniac to describe players who play extremely loosely and aggressively in all situations, often playing very weak hands very aggressively and putting in extra bets and raises when it is dearly incorrect to do so. It should be noted that players who play very well are often mistaken for maniacs, especially in short-handed or very tight games, because they too play weak hands aggressively and tenaciously. Exploiting these players often requires identifying how they respond to counter-aggression. If they back off when raised, then it is often correct to simply call with fairly strong hands, enticing them to bluff as many times as possible. On the other hand, if they ignore aggression or simply continue to bash away, then pushing strong hands is the proper exploitive response.
- ***Rocks*** - We use the term rock to describe players who play extremely tightly and often quite passively, entering the pot only with very strong hands and generally failing to extract value from those hands. Rocks rarely bluff, preferring the safety of their strong hands. Exploiting rocks is essentially about stealing their blinds and stealing pots on the flop when both players miss.
- ***Calling Stations*** - This term denotes a player who plays loosely but passively, often calling bets on many or all streets with weak hands or weak draws. Calling stations share the tenacity of a strong player, but fail to extract value from their hands appropriately. Exploiting calling stations is normally accomplished by value betting additional hands that are substantially weaker than the normal betting ranges, because these bets will now gain value against the expanded calling range of the calling station.

The reason that these strategies are so easy to spot and characterize is that they reflect rather quickly in the primary sources of information we discussed earlier. While players who raise 8% of their hands in a given spot are virtually indistinguishable from players who raise 12% of their hands in that spot after 100 hands, it is easy to identify a player who raises, for example, 50% of

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his hands. And in the case of a player who raises just ***one*** hand out of hundred, gets to showdown with it, and it happens to be a wired pair of aces, we can use Bayes' theorem to make a conclusion about that player's tendencies. Obviously, players play various strategies to degrees; the extreme maniac might simply be a raise-bot, who simply raises as many chips as he can at every opportunity. Such a player is relatively easy to beat. The more that the maniac tempers his aggression in spots where he is clearly the underdog, the closer he moves to playing well, the more difficult he Is to exploit, and the less valuable that exploitation is.

Caution should be used, though, when we make a read that a player plays poorly based on one or two examples. There are many dangers inherent in this process. First, the player may have simply made an uncharacteristic bad play. Poker is a complex game, and all players make mistakes of one form or another. Second, the player may have improved from when you first saw the weak play. This occurs frequently, especially in situations such as online where we might make a "note" about a play we saw. Months later, attempting to exploit the player who may have improved a lot in that time is dangerous. Third, the play may not be as weak as you believe—this occurs often when people are faced with loose, aggressive play that is actually quite strong but does not conform to conventional ideas about preflop "tightness." By automatically characterizing loose aggressive players as weak, many players ignore a severe danger that costs them significantly, as well as setting themselves up for frustration at their inability to beat these "weak" players.

The weaker the game, the more that players fall into easily exploitable categories. It is primarily in weak games that we think that the process of reading opponents' strategies pays off the most by quickly identifying and exploiting players who play very poorly. In the case of players who play fairly well, without any very large leaks, we often consider it to be more efficient and profitable to spend energy on identifying leaks hi our own play and preventing ethers from exploiting us.

Key Concepts

- Maximizing EV by playing exploitively requires that we formulate accurate and useful information about the opponents' distributions of hands and their strategies for playing those hands.
- *Some evidence we acquire is direct evidence-a hand revealed at showdown, for example.* This evidence is extremely valuable because it is known to be true.
- Indirect evidence, such as action frequencies, can also be a basis for inferences about play and strategy.
- Reading tells is essentially a Bayesian process-the value of a tell is related directly to both its frequency and reliability. False positives reduce the value of a tell substantially.
- Exploitive play reaches its maximum effectiveness against players who play very poorly.
- Our ultimate goal in hand-reading is to incorporate all the information we can gather to find the probability distribution of hands that our opponent holds.

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